



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Name: Shaik Nuha

Reg.no:192472242

Course Code: DSA0404

Course: Fundamentals of Data Science

Slot A

1. A feature "Age" ranges from 18–75 and another feature "Income" ranges from ₹20,000–₹20,00,000 in a dataset meant for a distance-based algorithm. Explain, with reasoning (not just definitions), why failing to standardize these variables would distort analysis, and calculate the z-score for Age = 45 if mean = 40 and SD = 12.

Given:

- Age ranges from 18 to 75
- Income ranges from ₹20,000 to ₹20,00,000
- The dataset is used for a distance-based algorithm.

Explanation

Distance-based algorithms, such as K-Nearest Neighbours (KNN) and K-Means, calculate distances between data points. If the variables are measured on very different scales, the variable with the larger numerical values can dominate the distance calculation.

For example, Age has values roughly between 18 and 75, whereas Income can range from ₹20,000 to ₹20,00,000. Therefore, the numerical difference in Income can be thousands or millions of units, while the difference in Age is only a few units.

As a result, without standardization, the algorithm would give much greater importance to Income than Age. This could distort distances, clusters, or nearest-neighbor results, even if Age is equally important.

Standardization puts the variables on a comparable scale, usually with mean 0 and standard deviation 1.

Z-score Formula

$Z = (X - \text{Mean}) / \text{Standard Deviation}$

For Age = 45:

Mean = 40

SD = 12

$$\begin{aligned} Z &= (45 - 40) / 12 \\ &= 5 / 12 = 0.4167 \end{aligned}$$

Therefore:

$$\mathbf{Z\text{-}score = 0.417}$$

This means that Age 45 is approximately 0.42 standard deviations above the mean.

2.A dataset has Mean = 45, Median = 45, Mode = 45, and skewness ≈ 0 , but kurtosis = 5.8 (leptokurtic). Explain what this combination tells you about the distribution's shape that skewness and central tendency alone would miss.

Given:

- Mean = 45
- Median = 45
- Mode = 45
- Skewness ≈ 0
- Kurtosis = 5.8

Interpretation

Since:

$$\text{Mean} = \text{Median} = \text{Mode} = 45$$

and

Skewness ≈ 0 ,

the distribution is approximately symmetric around 45. There is no significant tendency for the distribution to be pulled toward either the left or right side.

However, the kurtosis is 5.8, which indicates a leptokurtic distribution.

A leptokurtic distribution has:

- A relatively sharper or more concentrated peak.
- Heavier tails than a normal distribution.
- A greater tendency to contain extreme observations.

Therefore, skewness alone would suggest that the distribution is symmetric, but it would not tell us about the heaviness of the tails.

Conclusion

The distribution is approximately symmetric and centered around 45, but it has heavy tails and a higher likelihood of extreme values compared with a normal distribution.

Thus, central tendency and skewness describe the center and symmetry, while kurtosis provides additional information about tail heaviness and extreme observations.

3. Grouped Data — Mean and Variance

Class Interval: 0–10, 10–20, 20–30, 30–40, 40–50

Frequency: 5, 8, 15, 12, 10

Calculate the mean and variance using the step-deviation method. Show the assumed midpoints and working.

Given Data

Class Interval	Frequency (f)
0–10	5
10–20	8
20–30	15
30–40	12
40–50	10

Total frequency:

$$N = 5 + 8 + 15 + 12 + 10$$

$$N = 50$$

Step 1: Find Midpoints

Midpoint formula:

$$x = (\text{Lower Limit} + \text{Upper Limit}) / 2$$

Therefore:

Class	f	Midpoint (x)
0–10	5	5
10–20	8	15
20–30	15	25
30–40	12	35
40–50	10	45

Take assumed mean:

$$A = 25$$

Class width:

$$h = 10$$

Step 2: Calculate d

Formula:

$$d = (x - A) / h$$

Class	f	x	d	fd	fd²
0–10	5	5	-2	-10	20
10–20	8	15	-1	-8	8
20–30	15	25	0	0	0
30–40	12	35	1	12	12
40–50	10	45	2	20	40
Total	50			14	80

Step 3: Mean

Step-deviation formula:

$$\text{Mean} = A + h(\Sigma fd / N)$$

Substituting:

$$\text{Mean} = 25 + 10(14 / 50)$$

$$=25 + 2.8 = 27.8$$

Therefore:

$$\text{Mean} = 27.8$$

Step 4: Variance

Step-deviation formula:

$$\text{Variance} = h^2 [\Sigma fd^2 / N - (\Sigma fd / N)^2]$$

Substituting:

$$\text{Variance} = 10^2 [80/50 - (14/50)^2]$$

$$=100 [1.6 - 0.0784]$$

$$= 100 \times 1.5216$$

$$= 152.16$$

Therefore:

$$\text{Variance} = 152.16$$

Final Answer

$$\text{Mean} = 27.8$$

$$\text{Variance} = 152.16$$

4. Comparing Two Distributions via CV

Portfolio A: Mean return = 12%, SD = 3%

Portfolio B: Mean return = 18%, SD = 5%

An investor claims Portfolio B is "riskier in absolute terms but not necessarily worse per unit of return." Evaluate this claim quantitatively using Coefficient of Variation and explain what CV captures that SD alone doesn't.

Given:

Portfolio A

Mean return = 12%

SD = 3%

Portfolio B

Mean return = 18%

SD = 5%

Formula for Coefficient of Variation

$CV = (\text{Standard Deviation} / \text{Mean}) \times 100$

CV measures the amount of variability or risk relative to the mean return.

Portfolio A

$CV = (3 / 12) \times 100$

CV = 25%

Therefore:

CV of Portfolio A = 25%

Portfolio B

$CV = (5 / 18) \times 100$

CV = 27.78%

Therefore:

CV of Portfolio B = 27.78%

Comparison

Portfolio	Mean Return	SD	CV
A	12%	3%	25%
B	18%	5%	27.78%

Portfolio B has a higher standard deviation:

$5\% > 3\%$

Therefore, B has greater absolute variability or risk.

However, its CV is also higher:

$27.78\% > 25\%$

Therefore, Portfolio B has slightly more risk per unit of return as well.

Conclusion

The investor's statement is only partially correct.

Portfolio B is definitely riskier in absolute terms because its SD is higher.

However, when risk is compared relative to return using CV, Portfolio B is also slightly riskier:

- Portfolio A: $CV = 25\%$
- Portfolio B: $CV = 27.78\%$

CV captures relative variability, whereas SD measures absolute variability. This makes CV useful when comparing investments with different average returns or different scales.

5. Outlier's Effect on Variance

A dataset of 10 values has variance = 25. One value, originally 40 (close to the mean of 42), is mistakenly recorded as 400. Without recalculating exactly, explain directionally and reason about the magnitude of distortion to (a) the mean, (b) the variance — and why variance is affected disproportionately more than the mean.

Given:

- Number of values = 10
- Original variance = 25
- Original value = 40
- Mean ≈ 42
- Incorrectly recorded value = 400

The value 40 is close to the mean of 42, so originally it contributes very little to the variance.

However, recording it as 400 creates an extremely large deviation from the mean.

(a) Effect on Mean

Originally, the value is: 40

It is changed to: 400

The increase is:

$$400 - 40 = 360$$

Since there are 10 observations, the mean will increase substantially.

Approximately, the change in the mean is:

$$360 / 10 = 36$$

So the mean would increase from approximately 42 to roughly:

$$42 + 36 = 78$$

The exact new mean depends on the original values, but the important point is that the mean increases significantly.

(b) Effect on Variance

Variance is based on squared deviations from the mean.

Formula:

$$\text{Variance} = \Sigma(X - \text{Mean})^2 / N$$

Originally:

$$40 - 42 = -2$$

Squared deviation:

$$(-2)^2 = 4$$

So the original value contributes very little to variance.

After the error:

$$400 - \text{approximately } 42 = 358$$

Squared deviation:

$$358^2 \approx 128,164$$

This is enormously larger than the original squared deviation of 4.

Therefore, the variance will increase dramatically.

Why Is Variance Affected More Than the Mean?

The mean uses the original values directly, so the error increases the total by 360.

Variance uses squared deviations, so a very large deviation becomes even larger after squaring.

For example:

$$2^2 = 4$$

$$358^2 \approx 128,164$$

Therefore, a single extreme value can cause a massive increase in variance.

Conclusion

- **Mean:** Increases substantially.
- **Variance:** Increases extremely dramatically.
- **Reason:** Variance squares deviations, making it much more sensitive to extreme values than the mean.

Therefore, an incorrectly recorded extreme value such as 400 can seriously distort statistical analysis, especially variance and standard deviation.